Classical Mechanics Project

3D Rocket Trajectory with Atmospheric Drag and Coriolis Force

Rubén Praena, Claudia Ros, Miguel Sanz, Elishka Mrázek, Joel García and Giulia Latorre

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1. Context

In this exercise, we study the motion of a rocket launched from Earth, including:

- Altitude-dependent atmospheric density,
- Aerodynamic drag,
- Earth rotation (effective gravity and Coriolis effect),
- Variable mass due to fuel consumption.

The trajectory is computed numerically using Euler integration.

2. Physical Model

The rocket is subjected to the following forces, where GM = $3.99 \times 10^{1}4\ m^3s^{-3}$is the Earth's gravitational parameter, R = 6.378×10^6 m is the Earth's radius and z is the height above the Earth's surface, $\omega = 7.27 \times 10^{-5}$ rad/s is the Earth's spin velocity. For the rocket, T is the thrust, g_0 is the gravitational acceleration at the surface of the Earth. and I_{sp} is its specific impulse.

Gravity

$$\overline{g}(z) = \frac{-GM}{(R+z)^2}\ \hat{z}$$

Aerodynamic Drag

$$\overline{F}_D = -\frac{1}{2}C_d A\rho(z)v\overline{v}$$

Thrust (during burn phase)

$$\overline{F}_t = T\hat{z}$$

Coriolis Acceleration

$$\overline{a}_c = -2\overline{\omega} \times \overline{v}$$

Mass Variation

$$m' = -\frac{T}{g_0 I_{sp}}$$

3. Problem

3.1 Calculations

Consider a rocket characterised by the following parameters:

- m_0 = 100.0 kg, initial mass
- m_p = 10.0 kg, payload mass
- T= 1500.0 N, thrust
- I_{sp} = 100.0 s, specific impulse
- C_d = 0.5 drag coefficient
- A_{ref} = 0.1 m^2 cross-sectional area

Note: We used two models for the making of this project. The analytical solution neglects aerodynamic drag and therefore predicts a much larger burnout velocity and maximum altitude. However, the Euler numerical simulation includes atmospheric drag and density variation, producing much larger energy losses and a significantly lower maximum altitude. For each requested parameter, both models are shown and explained.

The rocket is launched vertically from the Earth's surface. Use Euler's method to determine:

a) The velocity $\vec{v}$ and position $\vec{r}$ of the rocket as a function of time.

Analytical approximation:

We launch vertically from the equator.

$\cdot\ \hat{x} = East$

$\cdot\ \hat{y} = North$

$\cdot\ \hat{z} = Up$

The rocket launches vertically, so initially:

$$\bar{v} = v\hat{z}$$

The earth rotates at the equator with an angular velocity:

$$\bar{\omega} = \omega\hat{y} = \frac{2\pi}{3600\cdot 24} = 7.2722\, rad/s\,\hat{y}$$

The rate of mass is:

$$m' = -\frac{T}{g_0 I_{sp}} \Rightarrow \int_{m_0}^{m(t)} dm = -\frac{T}{g_0 I_{sp}}\int_0^t dt$$

$$m(t) - m_o = -\frac{T}{g_0 I_{sp}} \cdot (t - 0)$$

$$\boxed{m(t) = m_o - \frac{T}{g_0 I_{sp}} \cdot t = m_0 - m' \cdot t} \quad (1)$$

Substituting values:

$$m' = -\frac{1500}{9{,}80665 \cdot 100} = 1.5296\, kg/s \quad | \quad m(t) = 100 - 1.5296 \cdot t$$

The propellant ends when $m(t) = m_p = 10\, kg$, so:

$$10 = 100 - 1.5296 \cdot t$$

$$t = \frac{-90}{-1{,}5296} = 58.8389\, s$$

Note: We assume the payload mass corresponds to the final rocket mass after propellant depletion.

Hence:

$$\boxed{m(t) = 100 - 1{,}5296 \cdot t \text{ for } t \in [0, 58.8389] s}$$

Now we need to know all the forces that are involved in the problem:

Gravity

$$\overline{g}(z) = \frac{-GM}{(R+z)^2}\, \hat{z}$$

Weight

$$\overline{F}_g = m(t) \cdot \overline{g}(z)$$

At $launch\ \overline{g}(0) = 9{,}8085\, \hat{z}\ m/s^2$

Aerodynamic drag:

$$\overline{F}_D = -\frac{1}{2} C_d A\, \rho(z) |\overline{v}|\, \overline{v}$$

The minus sign ensures $\overline{F}_D$ always opposes $\overline{v}$. The density $\rho(z)$ is given by the ISA[1] model (decreasing rapidly with altitude). At sea level: $\rho(0) = 1.225\, kg/m^3$

Thrust:

$$\overline{F}_z = T\hat{z} = 1500\, \hat{z}\ N,$$

Coriolis acceleration:

$$\overline{a}_c = -2\overline{\omega} \times \overline{v} = 2(\overline{\omega} \times \overline{v}) = 2(\omega\hat{y} \times (v_x\hat{x} + v_y\hat{y} + v_z\hat{z}))$$

[1] The ISA model, or International Standard Atmosphere, is a theoretical model that defines the standard atmospheric conditions at various altitudes, providing a reference for temperature, pressure, and density used primarily in aviation and meteorology.

$$\hat{y} \times \hat{x} = -\hat{z} \mid \hat{y} \times \hat{y} = 0 \mid \hat{y} \times \hat{z} = \hat{x}$$

$$\bar{a}_c = -2(\omega v_z \hat{x} - \omega v_x \hat{z}) = -2\omega v_z \hat{x} + 2\omega v_x \hat{z}$$

To find the velocity, we apply Newton's Second Law to each axis, obtaining the burn phase as a system of differential equations (2):

$$\bar{F} = m \cdot \bar{a} = m \cdot v' \Rightarrow \bar{v'} = \frac{\bar{F}}{m}$$

$$\boxed{\begin{aligned} v'_x &= -\frac{C_d A \rho(z)|\bar{v}|}{2m} v_x - 2\omega v_z \\ v'_y &= -\frac{C_d A \rho(z)|\bar{v}|}{2m} v_y \\ v'_z &= \frac{T}{m} - \frac{GM}{(R+z)^2} - \frac{C_d A \rho(z)|\bar{v}|}{2m} v_z + 2\omega v_x \end{aligned}}$$

These equations cannot be integrated analytically because $\rho(z)$ follows the piecewise ISA model (7 atmospheric layers) and z itself depends on v_z. We therefore proceed in two ways: analytically in the ideal limit, **ignoring Coriolis acceleration and the drag force,** and then numerically with Euler's method.

$$v'_z = \frac{T}{m(t)} - g_0 = \frac{T}{m_0 - m't} - g_0 \Rightarrow \int_0^t v_z' \, dv = \int_0^t \left(\frac{T}{m_0 - m't} - g_0\right) dt$$

Integrating the second half of the equation::

$$\boxed{v_z(t) = \int_0^t \frac{T}{m_0 - m't} \, dt - g_0 t}$$

Using u-sub:

$$u = m_0 - m't = m(t) \Rightarrow \text{eq. } 1$$

$$du = -m'dt; \; dt = \frac{-du}{m'}$$

$$\int_0^t \frac{T}{m_o - m't} \, dt = -\frac{T}{m'} \int_{m_0}^{m(t)} \frac{du}{u} = -\frac{T}{m'} (ln|m(t)| - ln|m_o|) = \frac{T}{m'} ln\left(\frac{m_0}{m(t)}\right)$$

Recall that: $\frac{T}{m'} = \frac{T g_0 I_{sp}}{T} = g_0 \, I_{sp}$

Then we obtain $v_z(t)$, that is Tsiolkovsky's equation plus the gravitational loss

$$\boxed{v_z(t) = g_o I_{sp} \, ln\left(\frac{m_0}{m(t)}\right) - g_o t}$$

At burnout, the time is $t_b = 58.8389 s$ and the mass is $m = 10 kg$

$$v_z(t_b) = 9,8067 \cdot 100 \cdot ln(\frac{100}{10}) - 9,8067 \cdot 58.8389 = 1681.0606 \, m/s$$

Integrating again to get z(t) z(t) during the burn. We need $\int_0^t v_z(t) \, dt$

$$\boxed{z(t) = g_o I_{sp} \int_0^t ln\left(\frac{m_0}{m_0 - m't}\right)dt - g_0 \int_0^t t\,dt}$$

Integrating the last term: $- g_0 \int_0^t t\,dt$

$$- g_0 \int_0^t t\,dt = - g_0 \cdot \frac{t^2}{2}$$

To integrate the first term: $g_o I_{sp} \int_0^t ln\left(\frac{m_0}{m_0 - m't}\right)dt$, we use the substitution: $u = m_0 - m't$

$$g_o I_{sp} \int_0^t ln\left(\frac{m_0}{m_0 - m't}\right)dt = g_o I_{sp} \int_{m_0}^{m(t)} ln\left(\frac{m_0}{u}\right) - \frac{du}{m'} = g_o I_{sp} \frac{1}{m'} \int_{m_0}^{m(t)} ln\left(\frac{m_0}{u}\right)du$$

Integrating by parts with $f = ln\left(\frac{m_0}{u}\right) du$, $dg = du$ so $df = -\frac{du}{u}$ and $g = u$.

$$ln\left(\frac{m_0}{u}\right) du = u\,ln\left(\frac{m_0}{u}\right) + u + C$$

Evaluating the bounds:

$$[m_0 \cdot ln(1) + m_0] - [m(t) \cdot ln\left(\frac{m_0}{m(t)}\right) + m(t)] = m't - m(t) \cdot ln\left(\frac{m_0}{m(t)}\right)$$

Using $m_0 - m(t) = m't$:

$$\frac{1}{m'} [m't - m(t) \cdot ln(\frac{m_0}{m(t)})] = t - \frac{m(t)}{m'} \cdot ln(\frac{m_0}{m(t)})$$

Now with the two terms integrated, we combine them:

$$z(t) = g_0 I_{sp} \cdot [t - \frac{m(t)}{m'} \cdot ln(\frac{m_0}{m(t)})] - \frac{g_0}{2} t^2$$

At burnout:

$$\frac{m(t_b)}{m'} = \frac{10}{1.5296} = 6.5376$$

$$z(t_b) = 980.67\,[58.8389 - 6,538 - 2,3026] - \frac{9.8067 \cdot 58.8389^2}{2} = 25962.6699\,m$$

At coast phase $t > t_b$ thrust and no drag:

$$v_z(t) = v_{z,bo} - g_0(t - t_b) = 1681.04 - 9.807(t - 58.8389)$$

$$z(t) = z_{bo} + v_{z,bo}(t - t_b) - \frac{g_0}{2}(t - t_0)^2$$

Using Euler's method:

To use Euler's method for a better approximation, we need to discretize the system of differential equations obtained above (2). Hence, we use a step of time small enough to make the results realistic: $\Delta t = 0.1s$.

From the ODE's, we declare a state vector s that includes all of the variables to be studied:

$$s = [x, y, z, v_x, v_y, v_z, m]$$

$$s' = f(s, t)$$

Euler's formula becomes:

$$\boxed{s^{n+1} = s^n + f(s^n, t^n)\Delta t}$$

For all seven variables:

$$x^{n+1} = x^n + v_x^n \Delta t$$

$$y^{n+1} = y^n + v_y^n \Delta t$$

$$z^{n+1} = z^n + v_z^n \Delta t$$

$$v_x^{n+1} = v_x^n + \left(-\frac{C_d A \rho(z^n)|\bar{v}|^n}{2m^n} v_x^n - 2\omega v_z^n\right)\Delta t$$

$$v_y^{n+1} = v_y^n + \left(-\frac{C_d A \rho(z^n)|\bar{v}^n|}{2m^n} v_y^n\right)\Delta t$$

$$v_z^{n+1} = v_z^n + \left(\frac{T}{m^n} - \frac{GM}{(R+z^n)^2} - \frac{C_d A \rho(z^n)|\bar{v}^n|}{2m^n} v_z^n + 2\omega v_x^n\right)\Delta t$$

$$m^{n+1} = m^n - m'\Delta t.$$

Step	$t\ [s]$	$z\ [m]$	$v_z\ [m/s]$	$m\ [kg]$	$a_z\ [m/s^2]$	$a_{x,Cor}\ [m/s^2]$
0 ⇒ 1	0	0	0	100.000	15.000-9.809 = 5.191	0
1 ⇒ 2	0.1	0	0.5191	99.847	15.023-9.809-0.0001 = 5.214	-7.55×10^{-5}
2 ⇒ 3	0.2	0.052	1.0406	99.694	15.046−9.809−0.0003 = 5.237	-1.51×10^{-4}
3 ⇒ 4	0.3	0.156	1.5643	99.541	15.069−9.809−0.0008 = 5.260	-2.27×10^{-4}
4 ⇒ 5	0.4	0.312	2.0903	99.388	15.092−9.809−0.0013 = 5.283	-3.04×10^{-4}

Table 1: approximation for the first 5 steps of the flight using Euler's method

b) The maximum height reached by the rocket

Analytical approximation:

Using the results from part 3.a:

After burnout, in the coast phase: $T = 0,\ m' = 0$, no drag:

$$v_z(t) = v_{z,bo} - g_o(t_{max} - t_b) = 0$$

$$t_{max} = t_b + \frac{v_{z,bo}}{g_o} = 58.8389 + \frac{1681.05}{9.807} = 230.2521\ s$$

To find z(t), or maximum height:

$$z_{max} = z_{bo} + v_{z,bo}(t_{max} - t_b) - \frac{1}{2}g_o(t_{max} - t_b)^2$$

We substitute the known values:

$$t_{max} - t_b = 171.41\ s$$

$$z_{max} = 25,958 + 1681.05 \cdot 171.41 - \frac{1}{2}9.8067(171.41)^2$$

$$z_{max} = 170,039.551m \simeq 170.0\ km$$

Using Euler's method:

After burnout (t_b=58.84 s), thrust becomes zero and the rocket enters the coast phase.

Euler's method is then continued using:

$$z^{n+1} = z^n + v_z^n \Delta t$$

$$v_z^{n+1} = v_z^n + a_z^n \Delta t$$

Where:

$$a_z^n = -\frac{GM}{(R+z^n)^2} - \frac{C_d A\rho(z^n)|\bar{v}^n|}{2m^n}v_z^n$$

The rocket reaches its maximum height when:

$$v_z = 0$$

Using the numerical simulation with $\Delta t = 0.1s$, the maximum altitude is approximately:

$$\boxed{z_{max} \simeq 1.09 \cdot 10^4 m}$$

Reached at:

$$\boxed{t_{max} = 66.6\ s}$$

The table below shows some steps indicating the main parameters in moments before the burnout and just after it.

$t\ [s]$	$z\ [m]$	$v_z\ [m/s]$	$m\ [kg]$	$a_z\ [m/s^2]$
58,6	9,846	349,79	10,367	10,79
58,7	9,881	350,87	10,214	10,87
58,8	9,916	351,96	10,061	10,94 before
58,9	9,951	353,05	10	-139,17 ← burnout (after)
59,0	9,986	339,14	10	-128,65

Table 2: Euler's approximation for the maximum flight reached, using the time of maximum flight

c) The time of flight t_F and the location of the fall.

Analytical approximation for time of flight:

We know the rocket lands when it returns $z = 0$

$$z(t_F) = 0; \quad t_F > 0$$

In the Euler simulation, z changes sign between two steps, we take the last step before $z < 0$.

Without drag, the flight has three phases, and the time $t_{max} = 230.2521$ and

$$z_{max} = 170,039.551 m$$

After the peak we know trust and drag are zero so: $z'' = -g_0$

$$z(t) = z_{max} - \frac{1}{2} g_0 (t_F - t_{max})^2$$

$$z(t_F) = 0 \Rightarrow 0 = z_{max} - \frac{1}{2} g_0 (t_F - t_{max})^2$$

$$\frac{1}{2} g_0 (t_F - t_{max})^2 = z_{max}; \quad (t_F - t_{max})^2 = \frac{2 \cdot z_{maz}}{g_0}$$

$$\Delta t_{fall} = \sqrt{\frac{2 \cdot z_{maz}}{g_0}} = 186.2209\, s$$

$$\Delta t_{fall} = t_F - t_{max} = 186.2209; \quad 186.22 + 230.2521 = 416.4730\, s$$

Analytical approximation for location of fall:

The rocket is launched vertically (v_x=v_y=0 at t=0). The resultant force in the x-axis, as we found previously is the following one:

$$v'_x = -\frac{C_d A \rho(z) |\bar{v}|}{2m} v_x - 2\omega v_z$$

But, as we neglect drag for the sake of simplicity, only the Coriolis component remains:

$$v'_x = -2\omega v_z \text{ and } v_y' = 0$$

From that we can know there is no North-South displacement, the only deviation is East-West.

Integrating to get $v_x(t)$:

$$\int_0^t v'_x dv = -2\omega \int_0^t v_z(t) dt$$

We find that $v_z = z'$ so

$$\int_0^t v_z(t)\, dt = \int_0^t z'\, dt = z(t) - z(0)$$

Finally we found the component of the position:

$$v_x(t) = -2\omega z(t)$$

Integrating to get $x(t_F)$

$$x(t_F) = \int_0^{t_F} v_x dt = -2\omega\int_0^{t_F} z(t)\, dt$$

The integral $\int_0^{t_F} z(t)\, dt$ has to be split on the different phases, burn, coast up and fall.

Phase 1, Burn: 0 to t_b, the z component can not be calculated analytically, so we obtain the value directly from Euler's method, in approximately 589 steps.

$$\int_0^{t_b} z(t)\, dt = \sum_{n=0}^{N_b-1} \frac{z^n+z^{n+1}}{2}\cdot \Delta t = \sum_{n=0}^{N_b-1} \frac{z^n+z^{n+1}}{2}\cdot 0.1 = 197549\, m/s$$

Phase 2, Coast up: t_b to t_{max}

$$z(t) = z_{bo} + v_{z,bo}(t - t_b) - \frac{g_0(t-t_b)^2}{2}$$

$$\text{Going to the integral, } \int_0^{t_F} z(t)\, dt \Rightarrow \int_0^{t_F} z_{bo} + v_{z,bo}(t - t_b) - \frac{g_0}{2}(t - t_b)^2\, dt$$

<u>First term:</u>

$$\int_{t_b}^{t_{max}} z_{bo}\, dt = z_{bo}(t_{max} - t_b)$$

<u>Second term</u>:

$$u = t - t_b\,, \quad du = dt$$

$$\int_{t_n}^{t_{max}} v_{z,bo}(t - t_b)\, dt = v_{z,bo}\int_0^{t_{max}-t_b} u\, du = v_{z,bo}[\frac{u^2}{2}]_0^{t_{max}-t_b} = \frac{1}{2}v_{z,bo}(t_{max} - t_b)^2$$

<u>Third term:</u>

$$u = t - tb$$

$$\int_{t_b}^{t_{max}} -\frac{g_0}{2}(t - t_b)^2\, dt = -\frac{g_0}{2}\int_0^{t_{max}-t_b} u^2\, du = -\frac{g_0}{2}[\frac{u^3}{3}]_0^{t_{max}-t_b} =-\frac{g_0}{6}(t_{max} - t_b)^3$$

With all the terms we found the outcome of the integral:

$$\int_0^{t_F} z_{bo} + v_{z,bo}(t - t_b) - \frac{g_0}{2}(t - t_b)^2\, dt = z_{bo}(t_{max} - t_b) + \frac{1}{2}v_{z,bo}(t_{max} - t_b)^2 - \frac{g_0}{6}(t_{max} - t_b)^3$$

$$25958 \cdot 171.41 + \frac{1681.05\cdot (171.41)^2}{2} - \frac{9,0867 \cdot (171.41)^3}{6} = 20913773\, m$$

Phase 3 Fall: t_{max} to t_F

$$\int_{t_{max}}^{t_F} z(t)\,dt = \int_{t_{max}}^{t_F} z_{max} - \frac{g_0}{2}(t - t_{max})^2 = \int_{t_{max}}^{t_F} z_{max} dt - \int_{t_{max}}^{t_F} \frac{g_0}{2}(t - t_{max})^2\,dt$$

$$\cdot \text{ Term 1: } \int_{t_{max}}^{t_F} z_{max}\,dt = z_{max}\cdot(t_F - t_{max})$$

$$\cdot \text{ Term 2 } (u = t - t_{max},\ \ du = dt):$$

$$\int_{t_{max}}^{t_F} \frac{g_0}{2}(t - t_{max})^2\,dt = \frac{g_0}{2}\int_0^{t_F - t - max} u^2\,du = \frac{g_0}{2}\left[\frac{u^3}{3}\right]_0^{t_F - t_{max}} = \frac{g_0}{6}(t_F - t_{max})^3$$

$$170037 \cdot 186.22 - \frac{9{,}0867 \cdot 186.22^3}{6} = 21109529$$

Now with al the phases calculated we can now all the value of $\int_0^{t_F} z(t)\,dt$

$$\int_0^{t_F} z(t)\,dt = 197,549 + 20,913,773 + 21,109,529 = 42,220,851\,m$$

Going to: $x(t_F) = -2\omega\int_0^{t_F} z(t)\,dt$

$$x_{land} = -2\cdot\frac{2\pi}{8.64\cdot 10^4}\cdot 42,220,851 = -6,138.9\,m$$

Using Euler's method for time of flight, $t\ (s)$, and location of fall, $x\ (m)$:
If we kept executing Euler's method, then we would find out that at approximately, the last steps of flight before z becomes zero are these, which occur around t=227s

$t\ (s)$	$z\ (m)$	$y\ (m)$	$x\ (m)$	$v_z\ (m/s)$	$v_x\ (m/s)$
227.2	28.0981	0	-3.0687	-57.1277	0.0492
227.3	22.3854	0	-3.0638	-57.1177	0.0492
227.4	16.6742	0	-3.0589	-57.0958	0.0492
227.5	10.9646	0	-3.0540	-57.0799	0.0491
227.6	5.2566	0	-3.0491	-57.0640	0.0491
227.7	-0.4498	0	**-3.0441**	-57.0481	0.0491 ← LAND

Table 3: Euler's approximation for the location and time of flight (in bold), x(m) and t(s) respectively. We also included the z and y coordinates for extra context.

Analytical calculations:

We define:

$$\lambda = \frac{-z_1}{z_2 - z_1} = \frac{-5.2566}{-0.4498-5.2566} = 0.9212$$

For landing time:

$$t_F = t_1 + \lambda(\Delta t)$$

$$t_F = 227.6 + 0.9212(0.1) \simeq 227.692\, s$$

For location of fall:

$$x_F = x_1 + \lambda(x_2 - x_1)$$

$$x_F =- 3.0491 + 0.9212(- 3.0441 + 3.0491) \simeq - 3.0445\, s$$

If we compare the results:

Quantity	Analytical (no drag)	Numerical (Euler + ISA)
t_F	$416.5\, s$	$227.7\, s$
x_{land}	$-6,138.9\, m$	$-3.044\, m$
y_{land}	0	0
Direction	West	West

3.2 Analysis

Using simulation outputs:

- **Plot altitude vs time.**

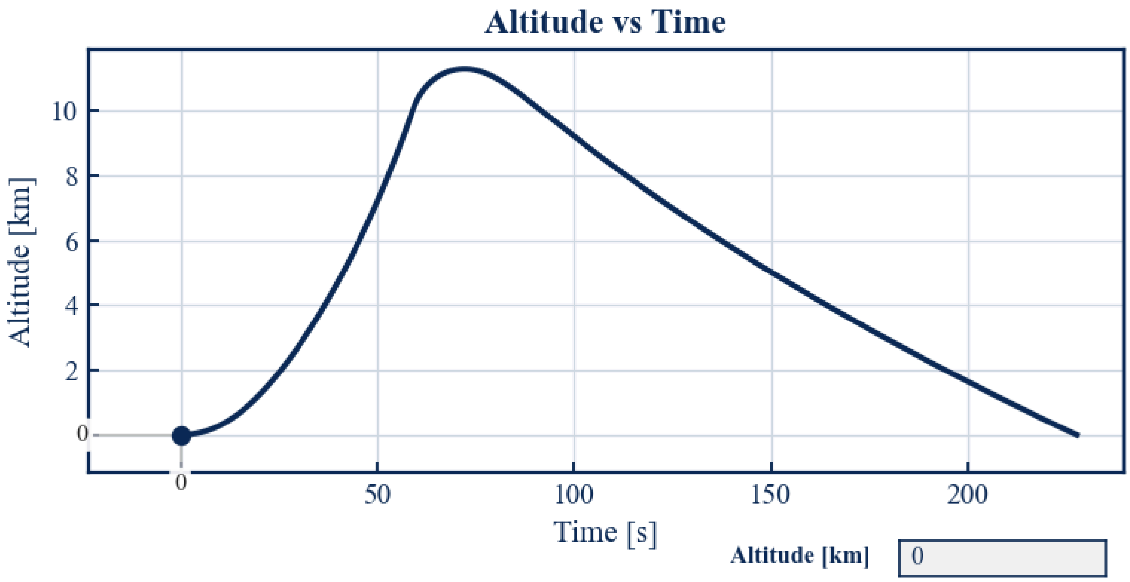


Figure 1: altitude vs time

- **Plot the horizontal projection of the trajectory (y versus x)**

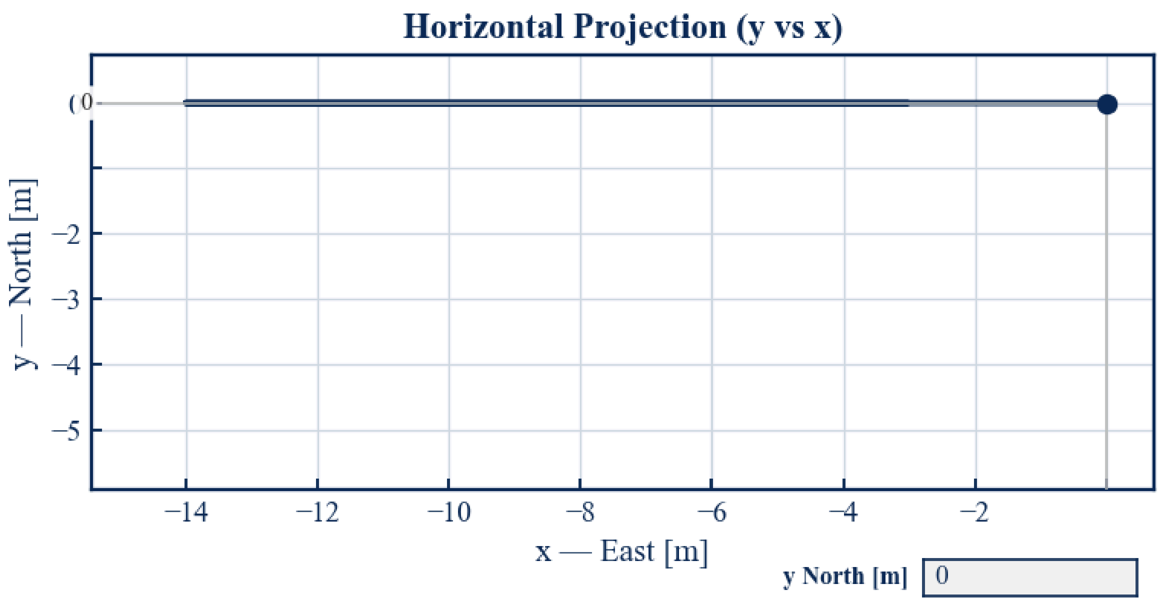


Figure 2: horizontal projection of the trajectory

- **Plot velocity vs time.**

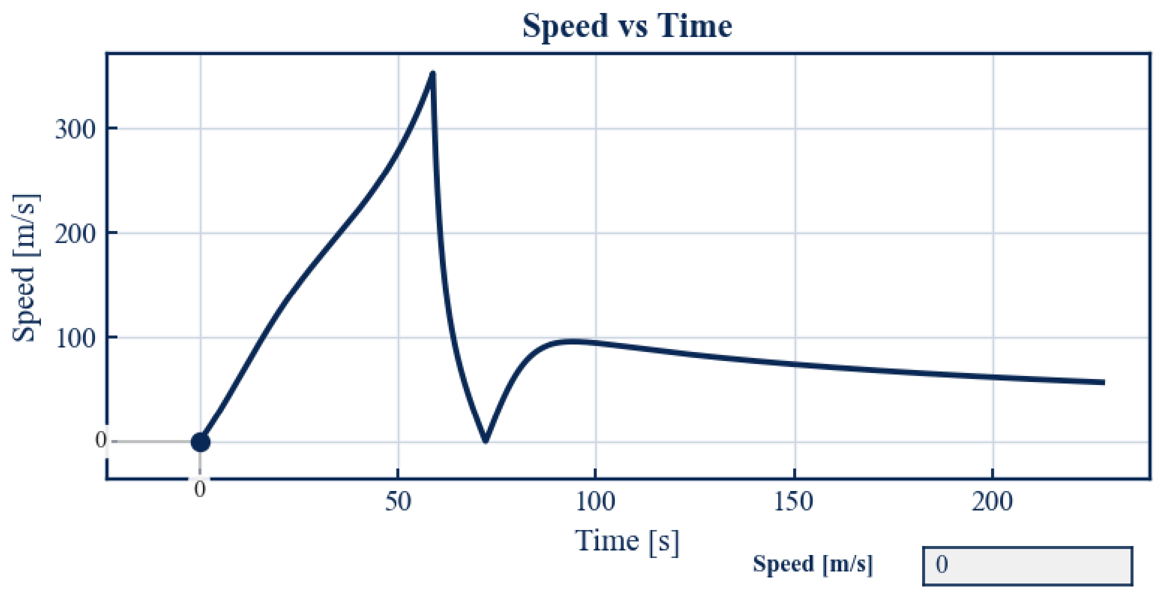


Figure 3: speed vs time

- **Plot drag vs time.**

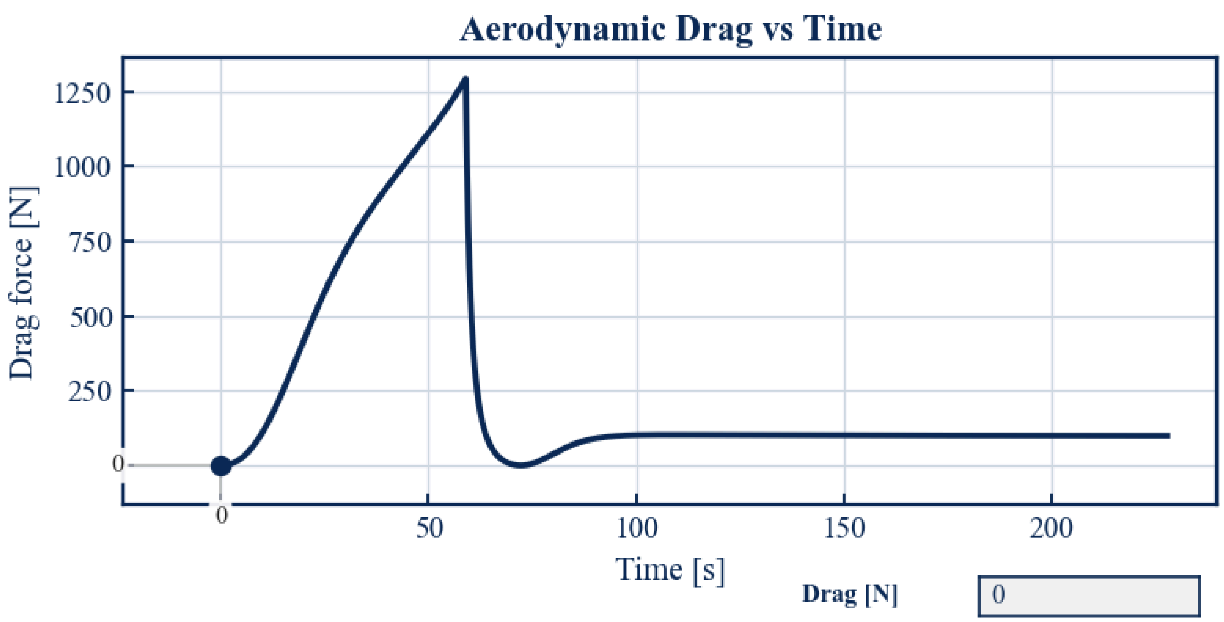


Figure 4: aerodynamic drag vs time

Answer the following questions:

d) Why does drag decrease rapidly with altitude?

Among other parameters, drag depends on the velocity of the body. As the rocket ascends, the atmospheric density (ρ) decreases exponentially according to the Standard Atmosphere (ISA) model. Because the aerodynamic drag force is directly proportional to this air density, this rapid drop in density causes the friction to plummet as we gain altitude, even though the rocket is moving really fast.

e) What happens to drag at very high altitude?

At very high altitudes, aerodynamic drag practically disappears and becomes equal to zero because the atmosphere becomes increasingly thinner. Beyond 85 km, in the exosphere region, air density (ρ) drops to such small values that there are almost no particles for the rocket to collide with. For this reason, even though the vehicle may be moving at extremely high speed, when an almost null density is introduced into the equation $\overline{F}_D = -\frac{1}{2}C_d A\rho(z)v\overline{v}$ the resulting drag force vanishes completely.

Hence , in the highest layers of the simulation, the rocket enters a free-flight stage in which air resistance no longer exists in the model, and the only relevant force acting on the object is gravity.

f) What is the range and the deviation of the rocket?

Based on our simulation results, the rocket experiences no North-South displacement ($y_{land} = 0$). The deviation only occurs along the East-West axis due to the Earth's rotation. In the analytical model (which ignores drag), the rocket drifts 6138.9 m to the West. However, in the numerical Euler simulation that includes atmospheric drag and density variations, this deviation is drastically reduced to a range of just 3.044 m to the West.

g) Which effect is more important, the Coriolis effect or friction?

Friction (aerodynamic drag) is much more important than the Coriolis effect in this specific trajectory. When we include atmospheric drag in our Euler simulation, it produces massive energy losses, reducing the maximum altitude from approximately 170 km (in the analytical model) to just 10.9 km. In contrast, the Coriolis acceleration only produces a minimal lateral drift of about 3 meters westward. Therefore, aerodynamic drag is by far the dominant factor and the main limitation in the rocket's performance.

h) What is the difference in the maximum height and range between the considered case, and a single-stage rocket such that the gas exhaust velocity is u = 1000 m/s and N = 10?

In our current study, the mass ratio is already $N = 10$ (since $m_o = 100\, kg$ and $m_p = 10\, kg$) , and the exhaust velocity is $u = g_o * I_{sp} \approx 980.67$. A theoretical single-stage rocket with $u = 1000\, m/s$ and $N = 10$ is extremely similar to our model but features a slightly more efficient exhaust velocity. According to Tsiolkovsky's rocket equation, this would yield a slightly higher ideal burnout velocity ($\Delta v = u * ln(N)$). Consequently, both the maximum height and the final westward deviation (range) would be slightly larger than the ones computed in our analytical approximation, though the overall order of magnitude would remain practically the same.

4. Additional computational Tasks

4.1 Task 1: Modify Thrust

Run simulations with T = 1000 N and T = 3000 N and answer the following questions:

Using simulation outputs:

- **Plot altitude vs time.**

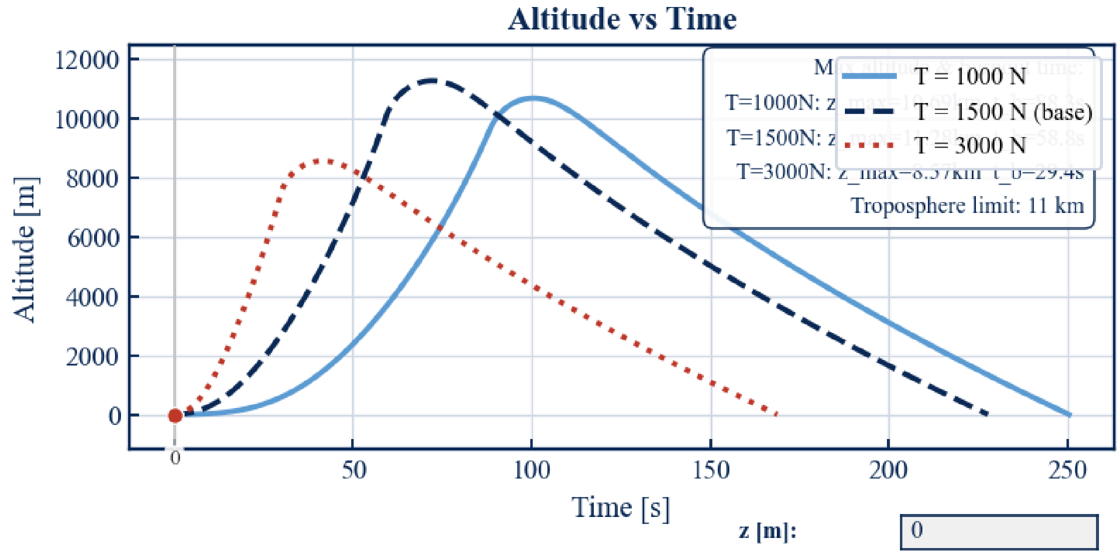


Figure 5: simulations varying thrust for altitude vs time

- **Plot speed vs time**

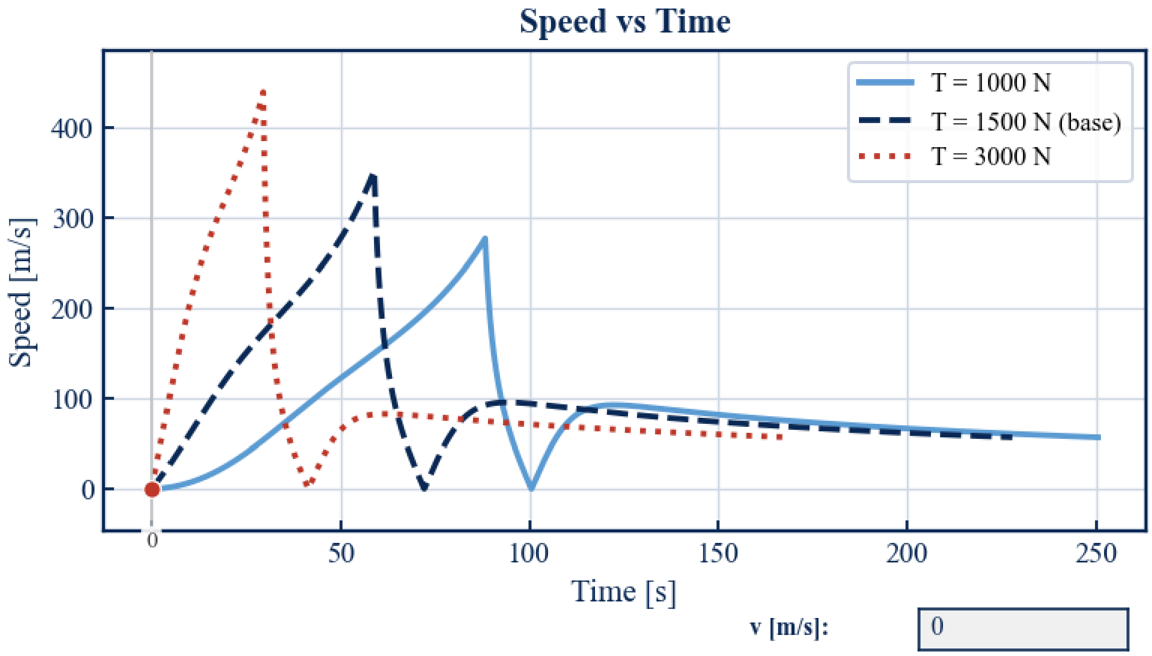


Figure 6: simulations varying thrust for speed vs time

- **Plot Aerodynamic Drag vs time.**
-

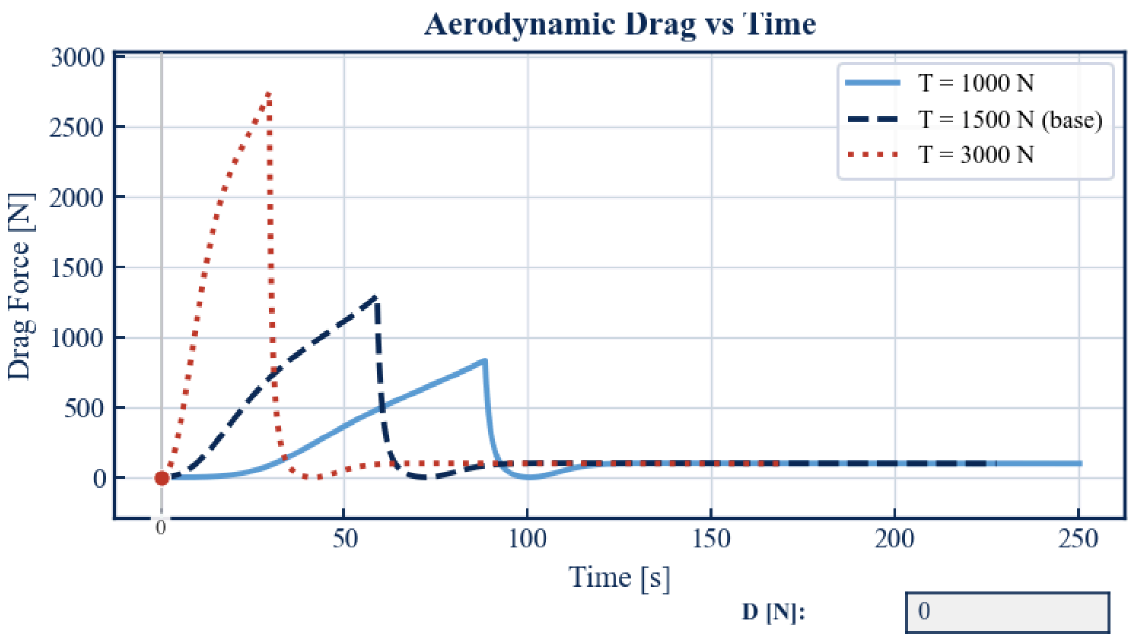


Figure 7: simulations varying thrust for drag vs time

- **Plot Mass vs time.**

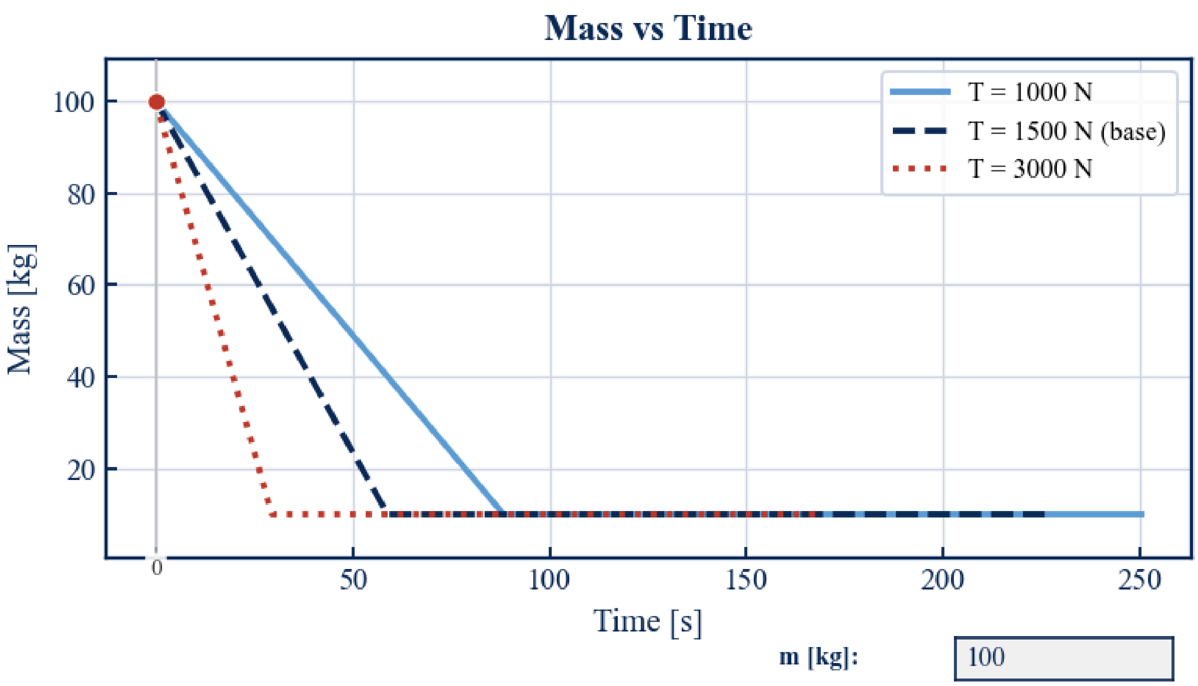


Figure 8: simulations varying thrust for mass vs time

- How does the maximum altitude change?

Looking at the altitude vs. time graph , increasing the thrust doesn't necessarily mean we will reach a higher altitude. The base case of $T = 1500\ N$ is actually the optimal setup here, reaching a peak altitude of about 11.3 km. If we reduce the thrust to $T = 1000\ N$, the rocket simply lacks the power and only reaches roughly 10.5 km. Surprisingly, if we blast the thrust up to $T = 3000\ N$, the maximum altitude drops significantly to just 8.57 km. This counterintuitive result happens because the high thrust accelerates the rocket to over 400 m/s very quickly. Flying that fast in the thickest part of the lower atmosphere creates a massive spike in aerodynamic drag (reaching almost 2700 N) , which ends up wasting a huge amount of energy.

- Does the rocket escape the dense atmosphere?

Barely, and it only happens in the base case. The upper limit of the dense troposphere is around 11 km. Only the $T = 1500\ N$ configuration manages to cross this boundary. Both the underpowered $T = 1000\ N$ case and the drag-heavy $T = 3000\ N$ case remain trapped within the troposphere.

- How does burnout time vary?

Assuming the specific impulse (I_{sp}) is a constant parameter for the engine, higher thrust directly increases the mass flow rate $m' = \frac{T_1}{g_o * I_{sp1}}$, meaning we burn through our fuel much faster. For $T = 3000\ N$, the fuel is depleted very quickly at around 30 seconds . Our base case ($T = 1500\ N$) burns out near 59 seconds , and the $T = 1000\ N$ configuration has the longest burn time, taking around 88 seconds to run out of propellant .

4.2 Task 2: Remove Coriolis Force

Run simulations with ω = 0 and answer the following questions:

Using simulation outputs:

- **Plot altitude vs time.**

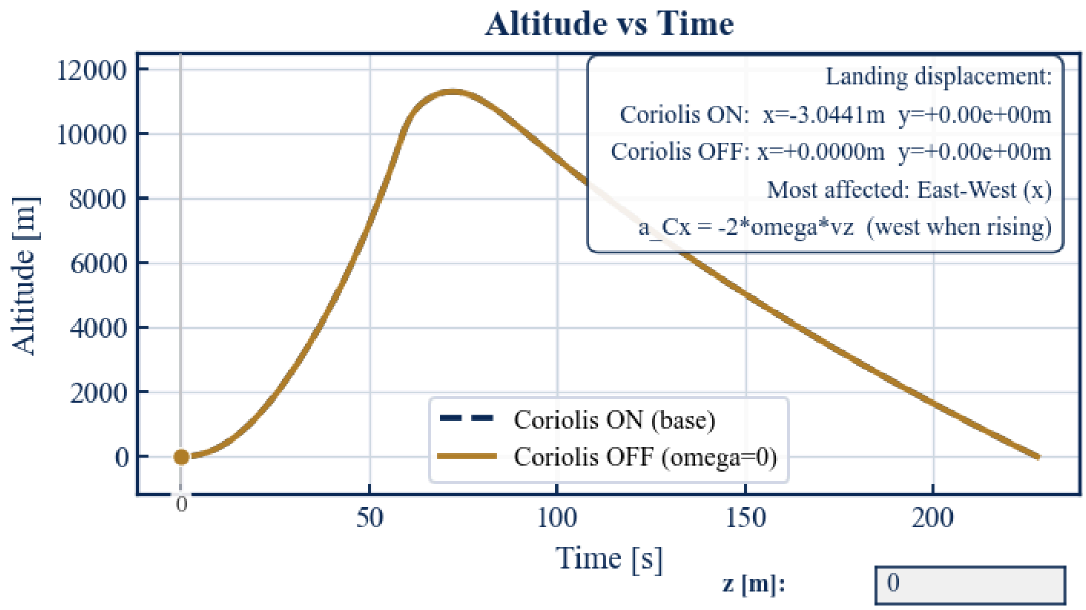


Figure 9: simulations removing coriolis force for altitude vs time

- **Plot East Displacement vs time.**

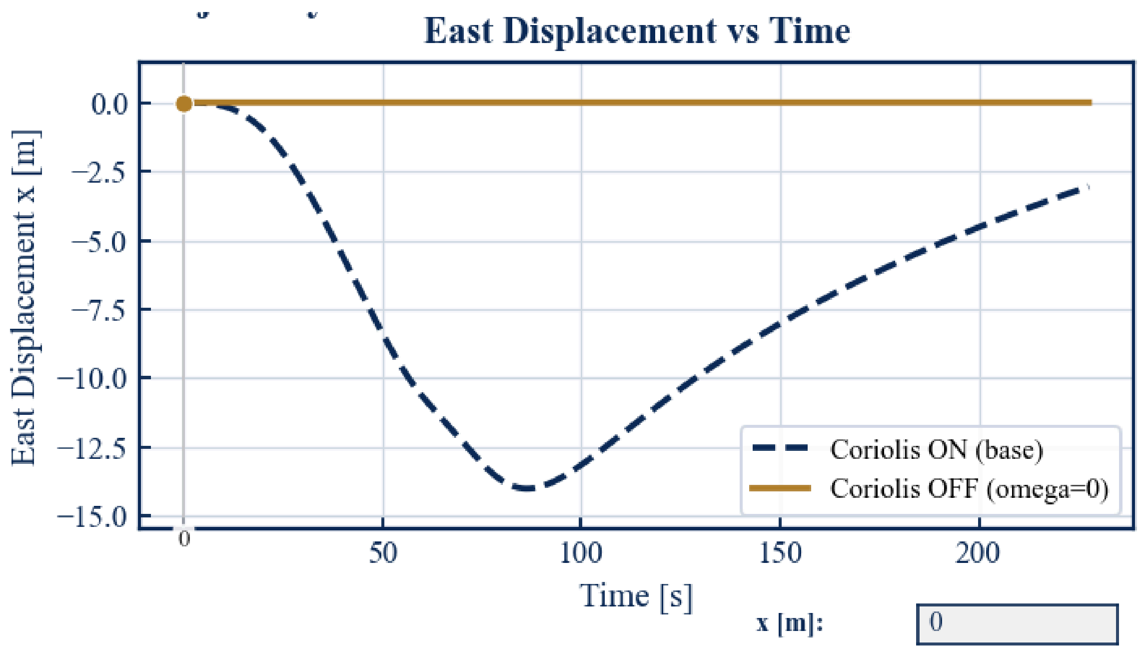


Figure 10: simulations removing coriolis force for East displacement vs time

- **Plot North Displacement vs time.**

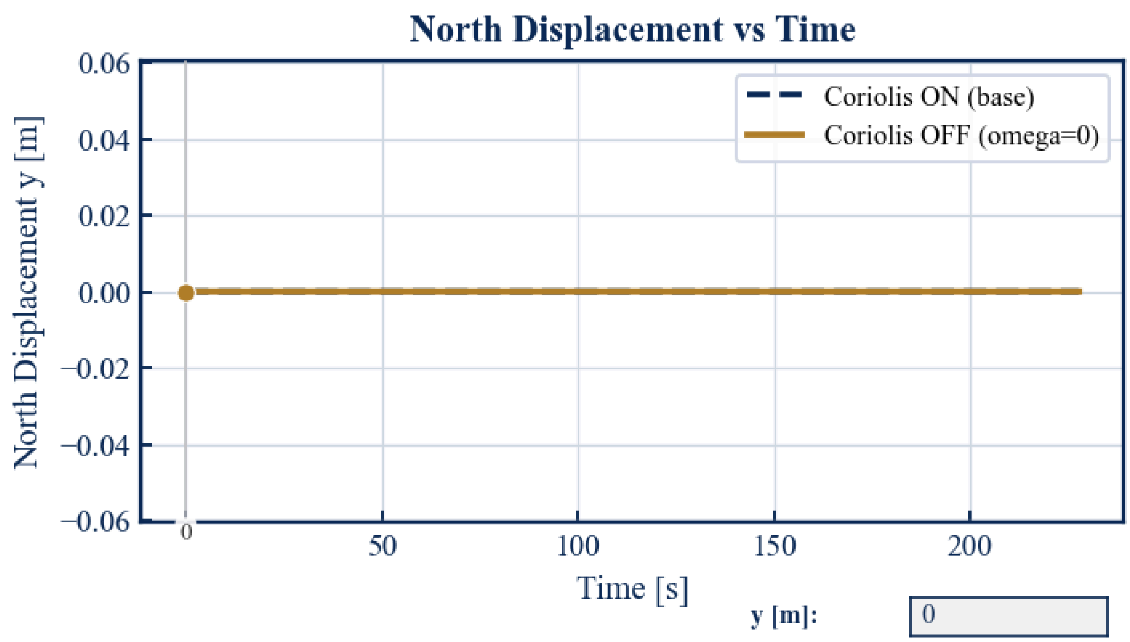


Figure 11: simulations removing coriolis force for North displacement vs time

- **Plot Horizontal Projection (y vs x).**

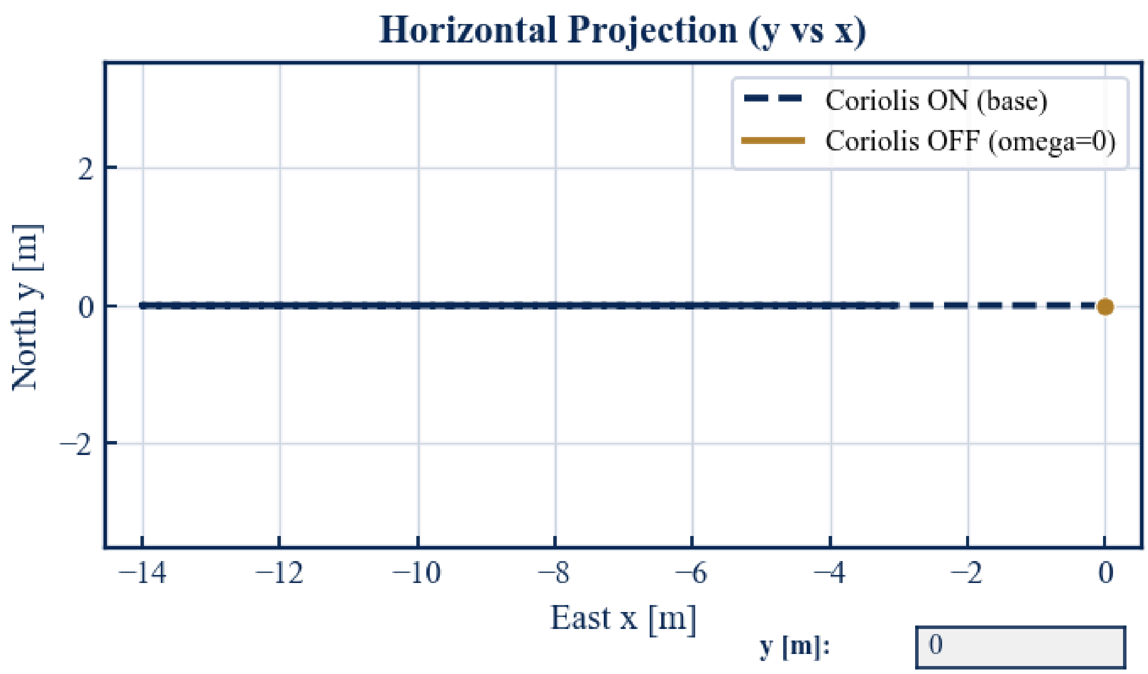


Figure 12: simulations removing coriolis force for horizontal Projection (y vs x)

<u>- How does the trajectory change?</u>

Removing the Coriolis force makes practically no difference to the vertical flight profile. The altitude vs time curves overlap perfectly. The only change is that the rocket stops drifting. With Coriolis ON, the rocket lands with a displacement of -3.0441 m, but with Coriolis OFF, it lands exactly at 0 m.

<u>- Which direction is most affected?</u>

The East-West (x-axis) direction is the only one affected by this force. The Coriolis acceleration on a vertical launch mainly acts horizontally because it is the cross product of the Earth's rotation vector and the rocket's vertical velocity ($a_{cx} = -2wv_z$). This pushes the rocket slightly to the West as it rises. The North-South (y-axis) displacement is completely unaffected and remains at zero .

4.3 Task 3: Simplified Atmosphere

Replace the International Standard Atmosphere (ISA) density with: $\rho(z) = \rho_0 e^{-z/H}$ and answer the following questions:

<u>Using simulation outputs</u>

- **Plot altitude vs time:**

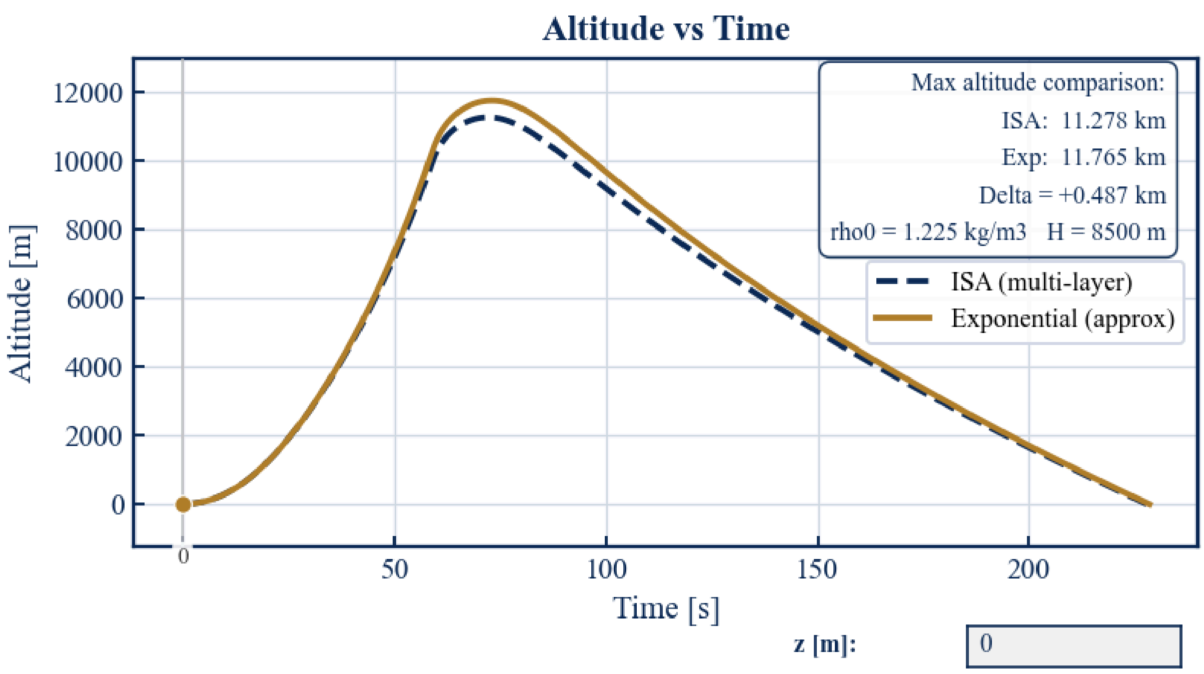


Figure 13: graph showing altitude vs time using python

- **Plot Aerodynamic Drag vs time:**

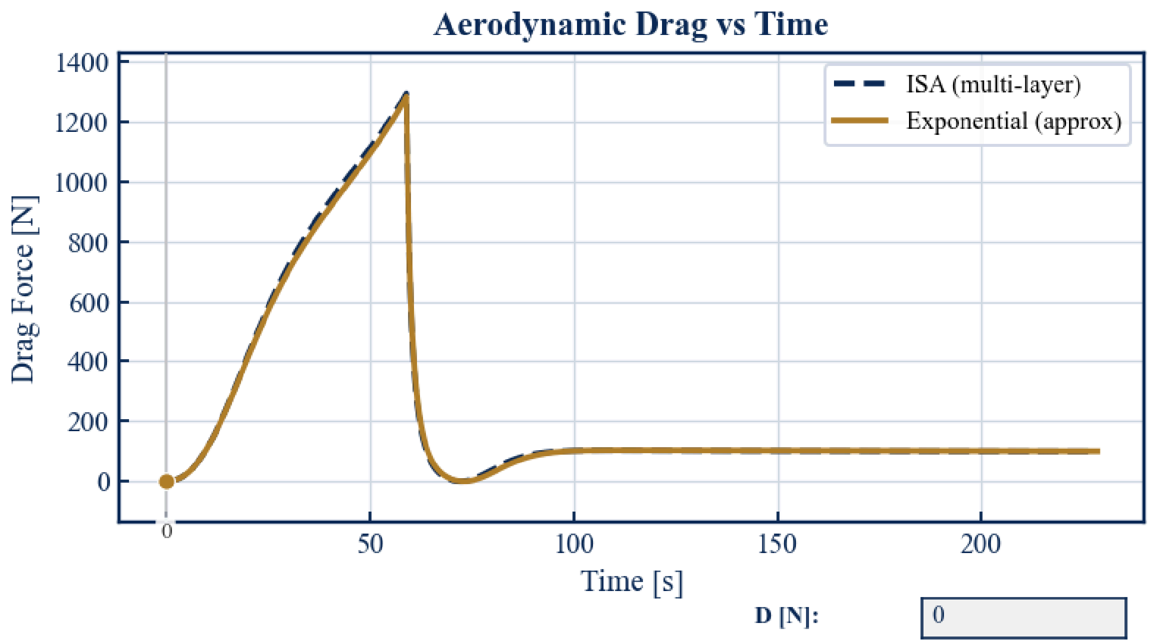


Figure 14: graph showing aerodynamic drag vs time using python

- **Plot speed vs time:**

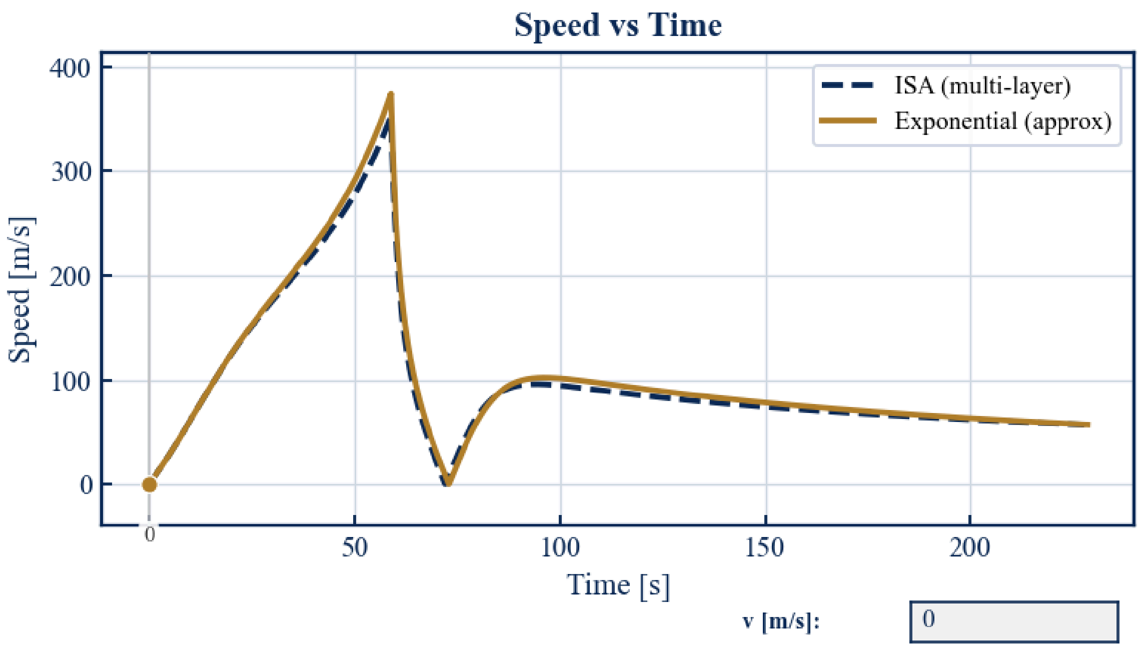


Figure 15: graph showing speed vs time using python

- **Plot air density experienced vs time:**

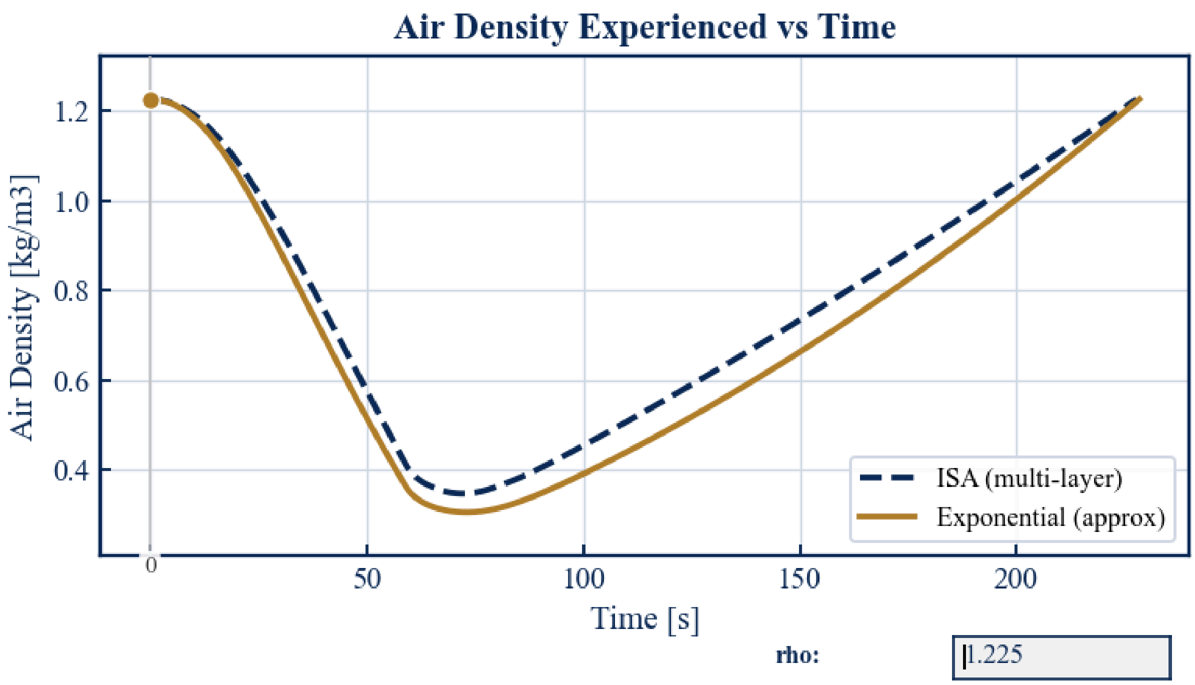


Figure 16: graph showing density experienced vs time using python

- Compare maximum altitude:

The simplified exponential model slightly overestimates the rocket's performance. It predicts a maximum altitude of 11.765 km, which is 0.487 km higher than the 11.278 km predicted by the multi-layer ISA model.

- Compare drag evolution:

While the drag curves follow the same general shape , the exponential model calculates a slightly lower air density in the upper layers compared to the ISA model. Because the computed density is lower, the aerodynamic drag is slightly underestimated, particularly around the period of peak velocity. This underestimation of drag is exactly why the exponential model predicts a higher maximum altitude.

- Which model is more realistic?

The ISA (multi-layer) model is definitively more realistic. The simplified exponential approximation assumes a constant temperature and scale height for the entire atmosphere. In reality, the ISA model accounts for actual temperature variations and lapse rates across different distinct atmospheric layers, providing a much more accurate density profile.

4.4 Task 4: Mass Variation

Change propellant mass 10 kg → 50 kg → 90 kg, and answer the following questions:

Using simulation outputs:

- **Plot altitude vs time:**

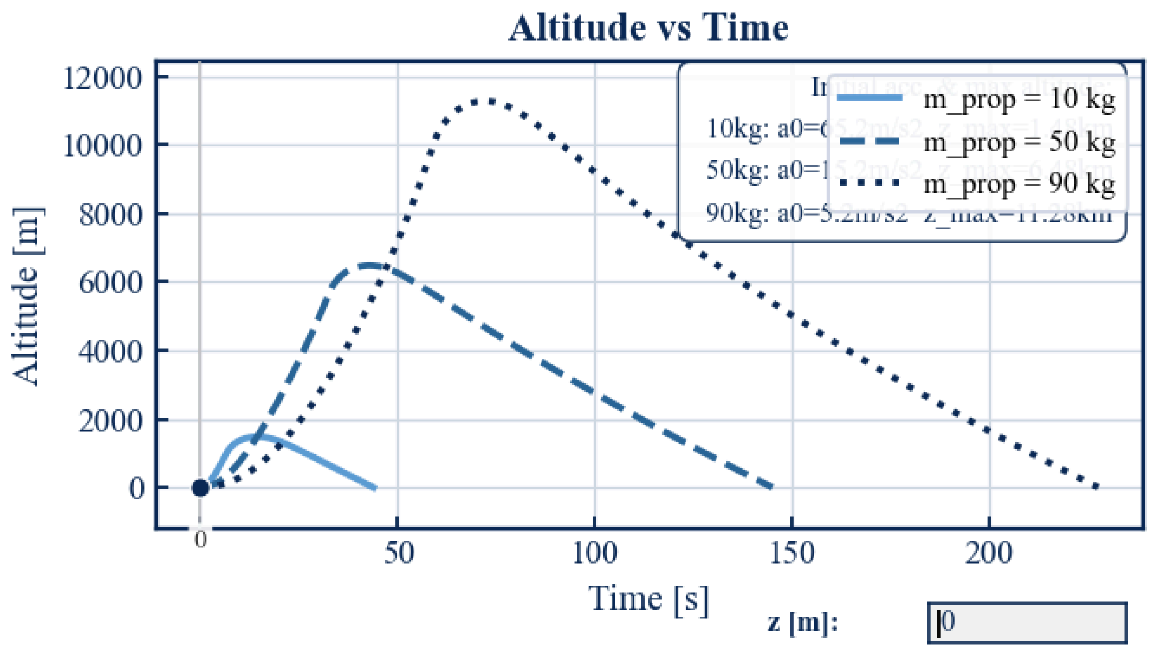


Figure 17: simulations with mass variation for altitude vs time

- **Plot speed vs time:**

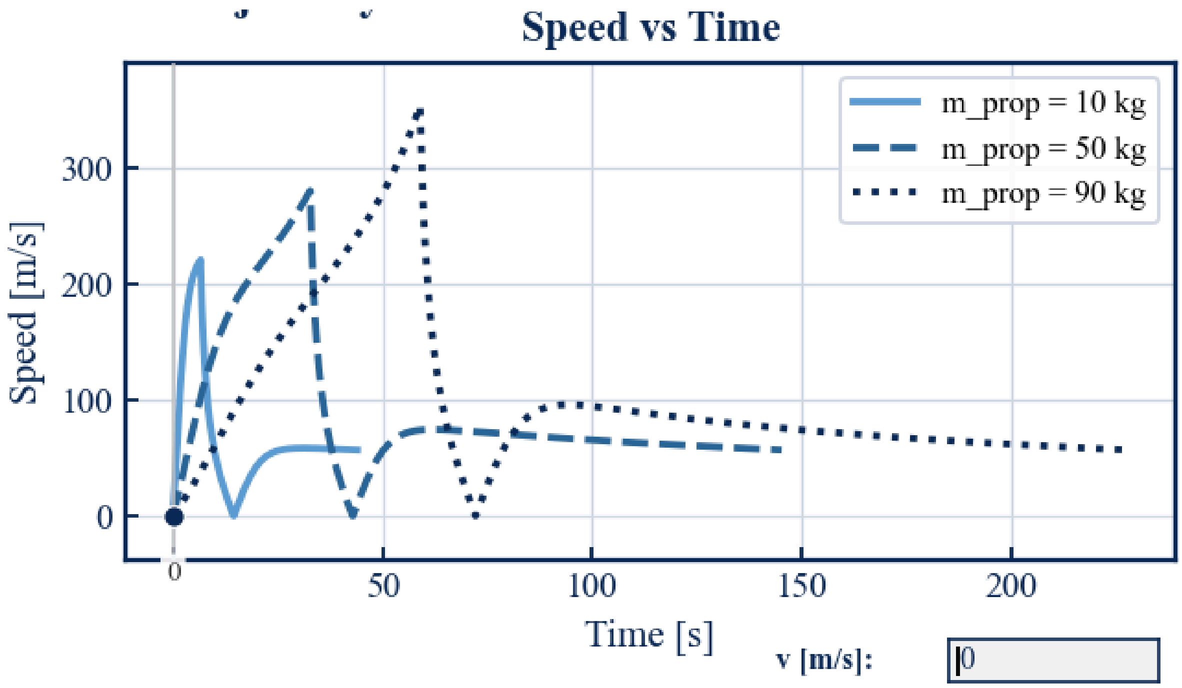


Figure 18: simulations with mass variation for speed vs time

- **Plot mass vs time:**

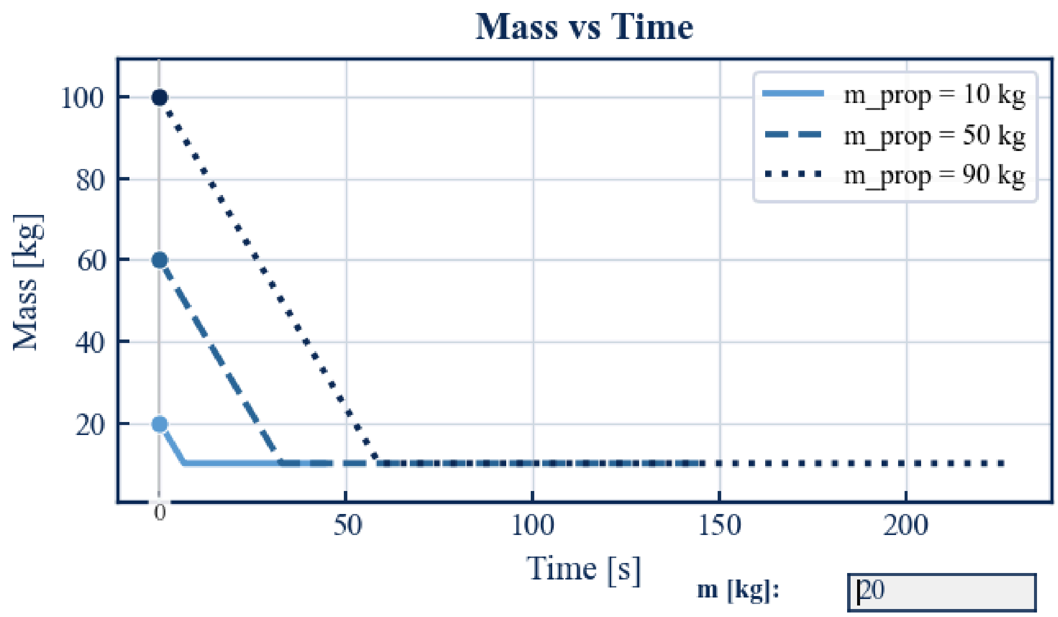


Figure 19: simulations with mass variation for mass vs time

- **Plot acceleration vs time:**

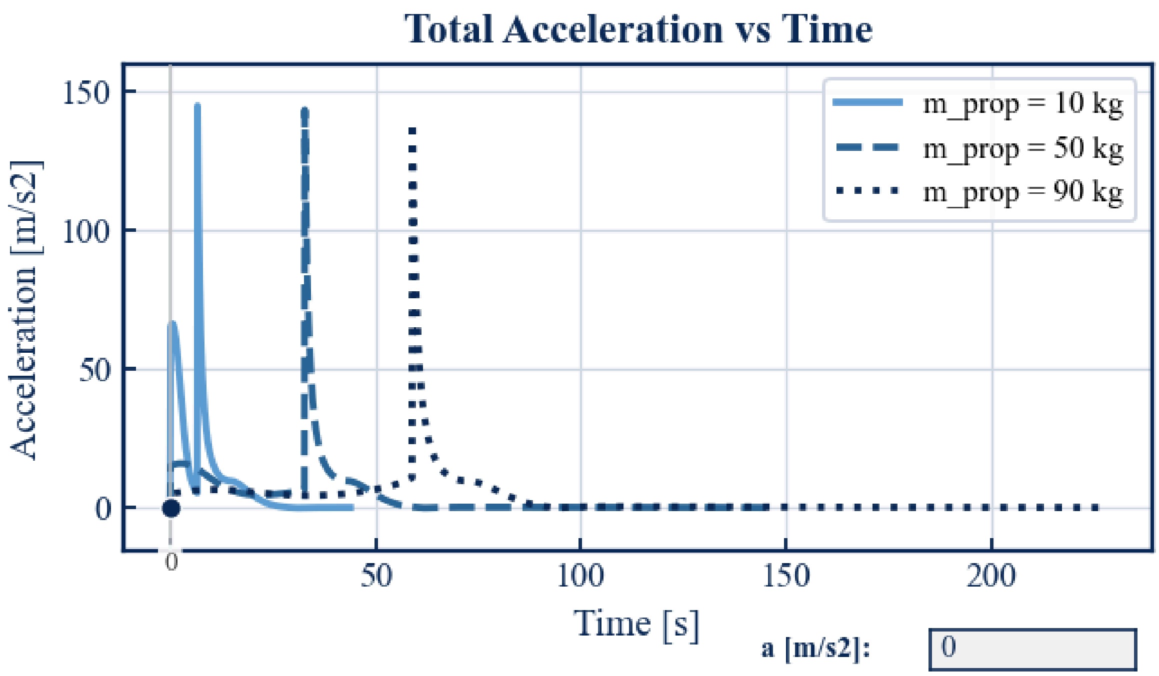


Figure 20: simulations with mass variation for total acceleration vs time

- <u>How does acceleration change?</u>

Because acceleration is inversely proportional to mass ($a = \frac{F}{m}$) , the initial acceleration is lower when the rocket carries more propellant. However, as the rocket burns fuel and loses mass, acceleration spikes exponentially right before burnout. The 10 kg propellant case has a massive and immediate acceleration spike (145 m/s^2) because the initial mass is already very low. The 90 kg case accelerates much more gradually, reaching its peak acceleration much later in the flight.

- <u>How does burnout altitude change?</u>

More propellant translates directly to a longer burn time, allowing the rocket to gain more velocity and reach a much higher altitude before the engine cuts off. The 10 kg propellant case burns out almost instantly at a very low altitude (roughly 1.5 km). The 50 kg case reaches approximately 6.5 km, while the 90 kg setup (our base case) reaches the highest burnout altitude, well over 10 km, before entering the coast phase.

5. Bonus exercise

Use the tools you have learned to think of a problem involving rockets and ballistic trajectories. Present the problem, solve it and analyse your results.

Introduction to our bonus activity:

For the bonus exercise, we are going to modify the single stage rocket into three separate stages. The first stage is going to maintain the initial structure, and will allow the rocket to reach 80km. This altitude will allow us to get through efficiently the parts of the atmosphere with higher density, and as a response, higher drag with a vertical launch. After Stage 1 separation, the second stage will ignite at high altitude, directing its thrust prograde to convert vertical speed into horizontal velocity until its burnout at approximately 171km. A brief unpowered coast phase then brings the rocket by inertia to the apogee of a sub-orbital arc at 265km, where Stage 3 burns prograde to circularise the orbit at a parking altitude of 266km. Finally, Stage 3 will execute a Hohmann transfer to climb towards 408km, performing a last circularisation burn at that estimated height to align, match speeds, and dock the rocket into the ISS.

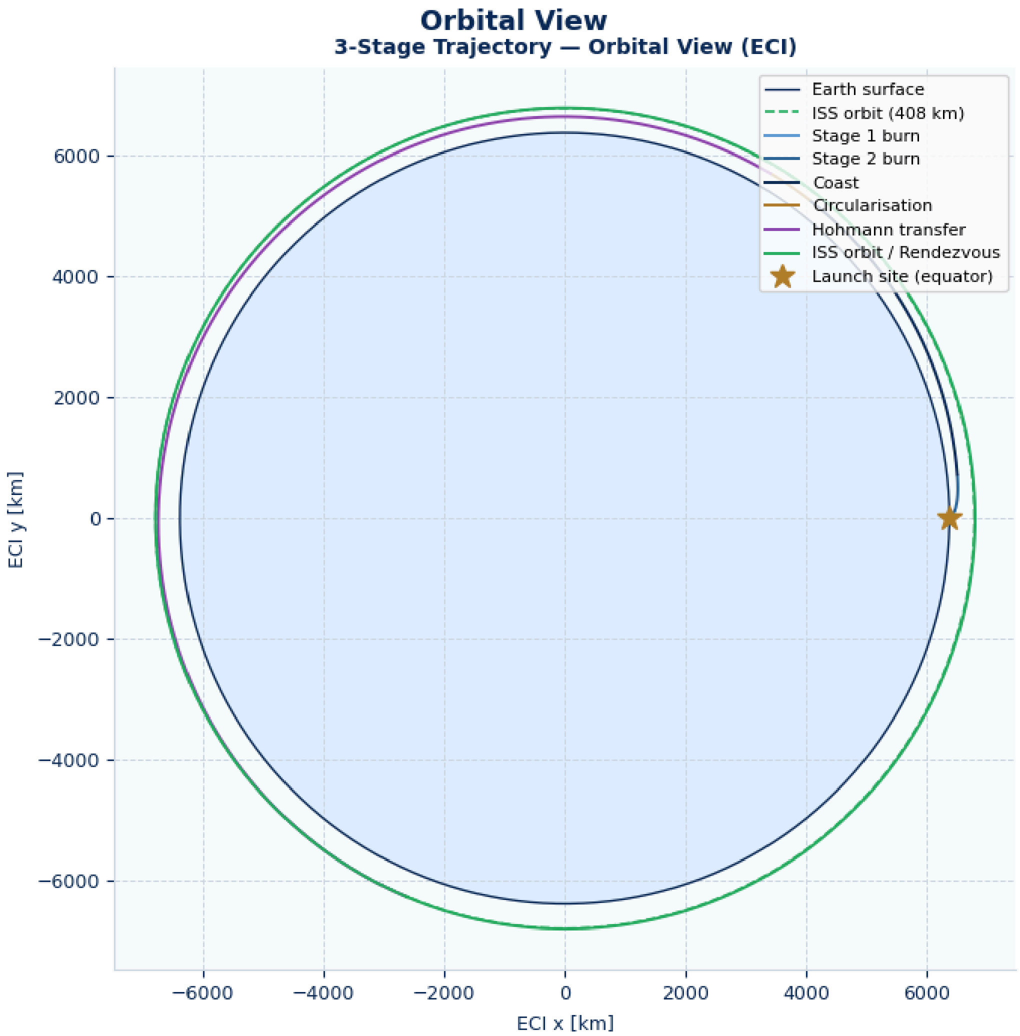


Figure 21: simulation of the orbital view of our 3-stage rocket trajectory

In order to accomplish this project we are going to use the Earth-Centered Inertial frame (ECI), that has its origin at Earth's center.

Parameters of each phase:

Stage	**Structural mass**	**Propellant mass**	**Thrust**	$\mathbf{I_{sp}}$
1	4,000 kg	40,000 kg	800kN	290s
2	1,000 kg	8,000 kg	200 kN	348 s
3	200 kg	500 kg	2 kN	300s
Payload	500 kg	—	—	—

Table 4: important parameters for each phase of the rocket

Once we have this parameters established, we are going to dive into the calculations of each phase and see if doing this is possible or not.

Following the same approach as Section 3.1 of the project, we now demonstrate the Euler integration by hand for the first steps of the mission.

The state vector is, once again:

$$s = [x, y, z, v_x, v_y, v_z, m]$$

The update rule for each variable is:

$$x^{n+1} = x^n + v_x^n \cdot \Delta t$$

$$y^{n+1} = y^n + v_y^n \cdot \Delta t$$

$$z^{n+1} = z^n + v_z^n \cdot \Delta t$$

$$v^{n+1}_z = v^n_z + [T/m^n - GM/(R+h)^2 - Cd\,A\rho(h)|v|v_z/(2m^n)] \cdot \Delta t$$

$$m^{n+1} = m^n - m' \cdot \Delta t,\; m' = T/(g_o \cdot I_{sp})$$

Analytical approximation:

<u>Stage 1</u>

Mass flow rate and burn time:

$$m'_1 = \frac{T_1}{g_o * I_{sp1}} = \frac{800x10^3}{9.8x290} = 281.301\ kg/s$$

The first stage propellant mass is 40,000 kg, so burn time:

$$t_{burn,1} = \frac{m_{prop}}{m'_1} = 142.20\ s$$

Mass as a function of time during Stage 1 burn:

$$m(t) = m_o - m'_1 \cdot t = 54\,200 - 281.301 \cdot t\ for\ t \in [0,\ 142.2]\ s$$

Using Euler's Method to prove that mass lifts the rocket:

For the first few steps h << R, so gravity ≈ −g 0 = −9.807 m/s² and density ρ ≈ ρ 0 = 1.225 kg/m³. The aerodynamic drag at t = 0 is zero (v = 0). The initial net vertical acceleration is:

$$a_{z,0} = T/m_0 - g_0 = 800,000 / 54,200 - 9.807 = 14.7601 - 9.807 = 4.9535\,m/s^2$$

$$\frac{T}{m_0} = 14.760\, m/s^2 > g_o$$

Hence, the rocket will lift off. The thrust-to-weight ratio at launch is:

$$\frac{T}{m_o g_o} = 1.505$$

Using Euler's Method to calculate the final velocity for the first stage

Using the same procedure as the one explained in part 3.1 and showcased in *Table 1*, we create the first Euler steps for stage 1:

Step	$t\ [s]$	$z\ [m]$	$v_z\ [m/s]$	$m\ [kg]$	$a_T\ [m/s^2]$	$a_{drag}\ [m/s^2]$	$a_{net}\ [m/s^2]$
0→1	0	0	0	54200.000	14.7601	0	4.9516
1 →2	0.1	0	0.4952	54171.870	14.7678	−0.0000	4.9593
2 →3	0.2	0.0495	0.9911	54143.740	14.7755	−0.0001	4.9689
3→4	0.3	0.1486	1.4878	54115.610	14.7832	−0.0001	4.9746
4→5	0.4	0.2974	1.9853	54087.480	14.7909	−0.0002	4.9822

Table 5: approximation for the first 5 steps of the first stage using Euler's method

Velocity at Burnout:

Continuing the Euler integration for all 1422 steps until m(t) = m_bo = 14,200 kg (t = 142.2 s), the numerical simulation gives:

$$v_{z,bo} = 2716.1\ m/s$$

Ignoring drag and using Tsiolkovsky (as in Section 3.1 of the main project) with a gravitational loss – $g_o\, t_{burn}$:

$$v_{z,bo} = g_0 I_{sp} ln(m_o / m_{bo}) - g_0 t_{burn} =$$

$$9.807 \times 290 \times ln(54\,200 / 14200) - 9.807 \times 142.2 \approx 2414.8\, m/s$$

The velocity using Tsiolkovsky equation is higher than the numerical simulation, which may seem counterintuitive since drag should reduce performance. The explanation lies in the

gravity turn: by pitching the thrust vector progressively toward prograde, the rocket reduces its gravitational loss. This efficiency gain outweighs the drag losses in the upper atmosphere, resulting in a net higher total speed at burnout.
In a purely vertical trajectory, the full gravitational loss is:

$g_o * t_{burn} = 9.807 \times 142.2 = 1394\ m/s$

The gravity turn recovers a significant fraction of this, which explains why the simulation outperforms the simple Tsiolkovsky estimate.

Stage 2

After Stage 1 separation, the remaining vehicle mass is $m_{2,O}$= 10200 kg. Stage 2 has T = 200,000 N and I_{sp} = 348 s:

Mass flow rate and burn time:

$$m'_2 = \frac{T_2}{g_o * I_{sp2}} = \frac{200x10^3}{9.8x348} = 58.604\ kg/s$$

The Stage 2 propellant mass is 8,000 kg, so burn time:

$$t_{burn,1} = \frac{m_{prop}}{m'_1} = 136.5\ s$$

Velocity at burnout:
Stage 2 operates at high altitude (>80 km) where atmospheric density is negligible, so drag losses are small. The thrust is directed prograde (tangential to the trajectory), converting vertical speed into the horizontal orbital velocity needed for a stable orbit.

$$\Delta V_{2,ideal} = 9.807 \times 348 \times ln(10200 / 2200) = 5234.9\ m/s$$

After Stage 2 burnout, the rocket is at approximately h = 171 km with a mostly horizontal velocity vector. A brief coast phase (using the inertia of the rocket) brings it to the apogee of a sub-orbital arc at h ≈ 265 km

Stage 3

Stage 3 burns prograde to circularise the orbit accomplished at stage 2. The required circular orbital velocity at the parking altitude (h park = 266 km) is:

$$v_{circ,park} = \sqrt{(GM/r_{park})} = 7749.5\ m/s$$

Mass flow rate:

$$m'_3 = \frac{T_3}{g_o * I_{sp3}} = \frac{2x10^3}{9.8x300} = 0.6803\ kg/s$$

Hohmann Transfer to ISS (h = 408 km):

A Hohmann transfer is the most fuel-efficient two-impulse manoeuvre to move between two circular coplanar orbits. It consists of two tangential burns:

- **Burn 1 at perigee of parking orbit (h = 266 km):** raises the apogee to the ISS altitude, placing the rocket on an elliptical transfer orbit.
- **Burn 2 at the apogee of the transfer orbit (h = 408 km):** circularises the orbit at the ISS altitude.

The semi-major axis of the Hohmann transfer ellipse is:

$$a_{transfer} = (r_{park} + r_{ISS}) / 2 = (6644.0 + 6786.0) / 2\ km = 6715.0\ km$$

Velocities on the transfer orbit at perigee and apogee:

$$v_{transfer,p} = \sqrt{GM \cdot (\frac{2}{r_{park}} - \frac{1}{a})} = 7790.32\ m/s$$

$$v_{transfer,a} = \sqrt{GM \cdot (\frac{2}{r_{ISS}} - \frac{1}{a})} = 7627.31\ m/s$$

Required velocity increment at the first burn, making an elliptic orbit:

$$\Delta v_{H1} = V_{transfer,p} - V_{circ,park} = 7790.32 - 7749.46 = 40.86\ m/s$$

Required velocity increment at the second burn, circularising the orbit at the ISS altitude :

$$\Delta v_{H2} = V_{circ,ISS} - V_{transfer,a} = 7667.95 - 7627.31 = 40.65\ m/s$$

And the total Hohmann velocity:

$$\Delta v_{Hohmann,total} = 40.86 + 40.65 = 81.51\ m/s$$

Burn time:

Each burn lasts only ~10 s and consumes ~7 kg of propellant.

ISS Orbital Period:

From Kepler's third law:

$$T_{ISS} = 92.7\ min = 5560.5\ s$$

All these calculations explain analytically what the code is simulating and plotting. The final graphs show exactly what our three-stage rocket is doing in every stage:

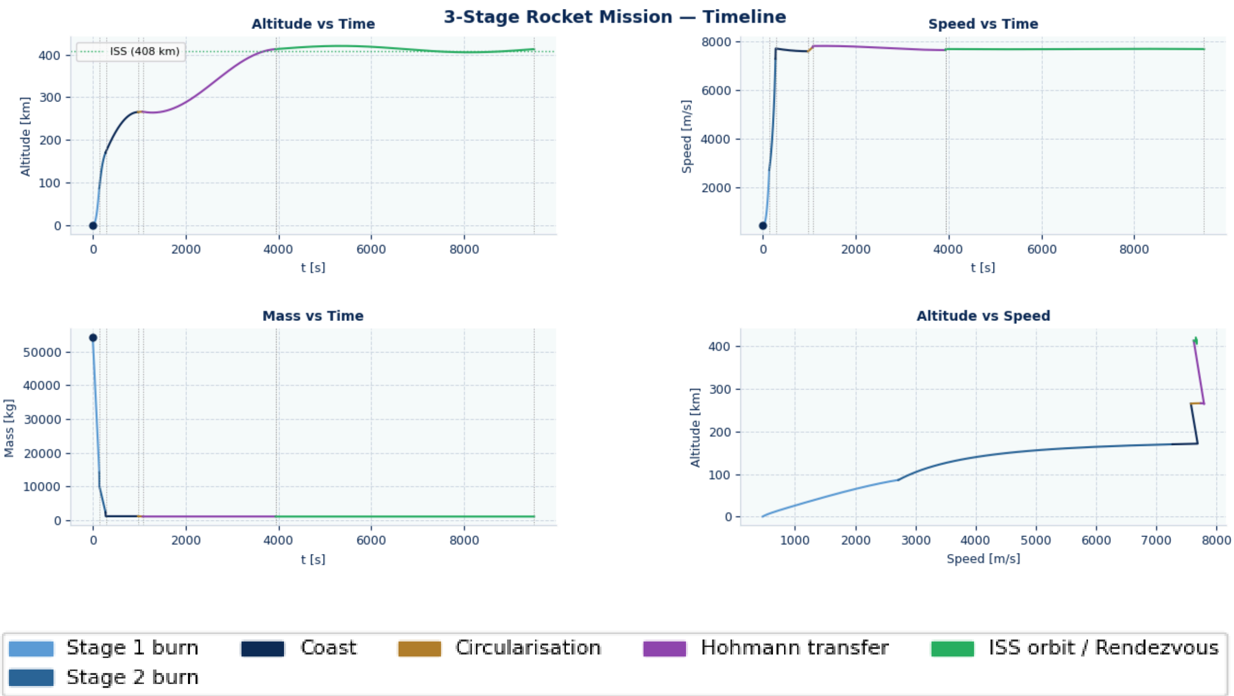


Figure 22: timeline of our 3-stage rocket mission, including the graphs: altitude vs time, speed vs time, mass vs time and altitude vs speed, shown below with more detail. Also, colour code for the following and current graphs is shown.

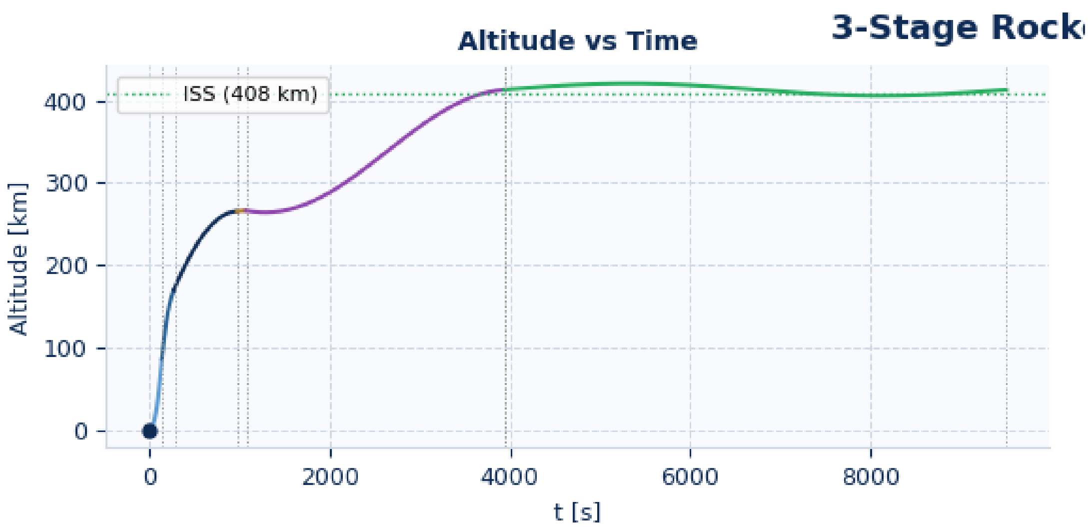
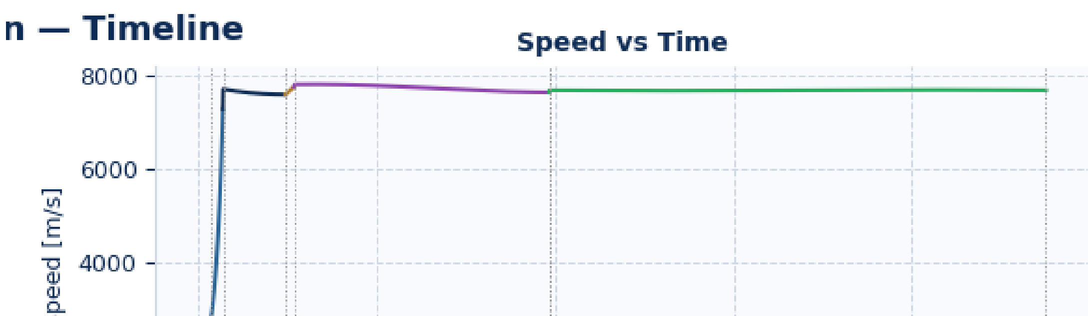


Figure 23: graph of altitude vs time for our rocket (above)

Figure 24: graph of speed vs time for our rocket (below)



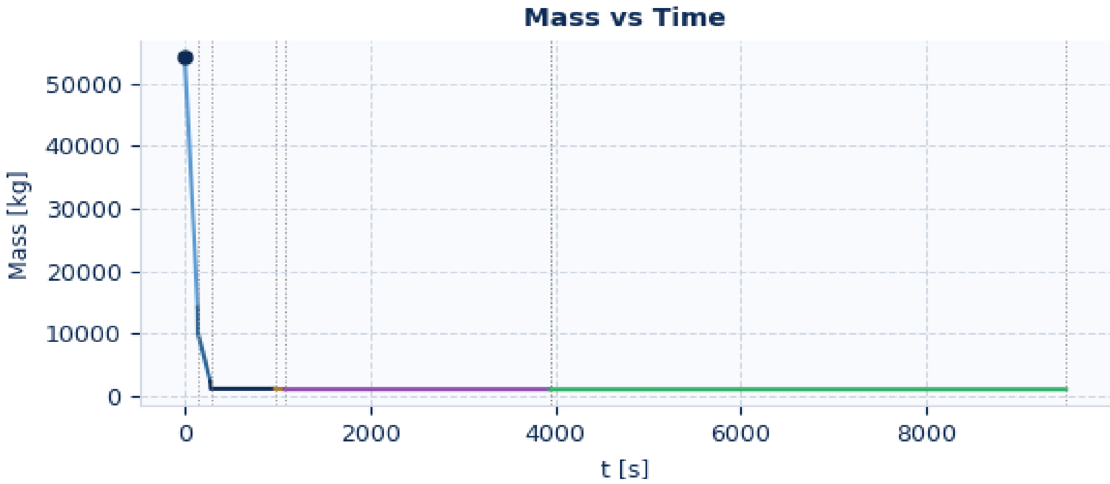


Figure 25: graph of mass vs time for our rocket

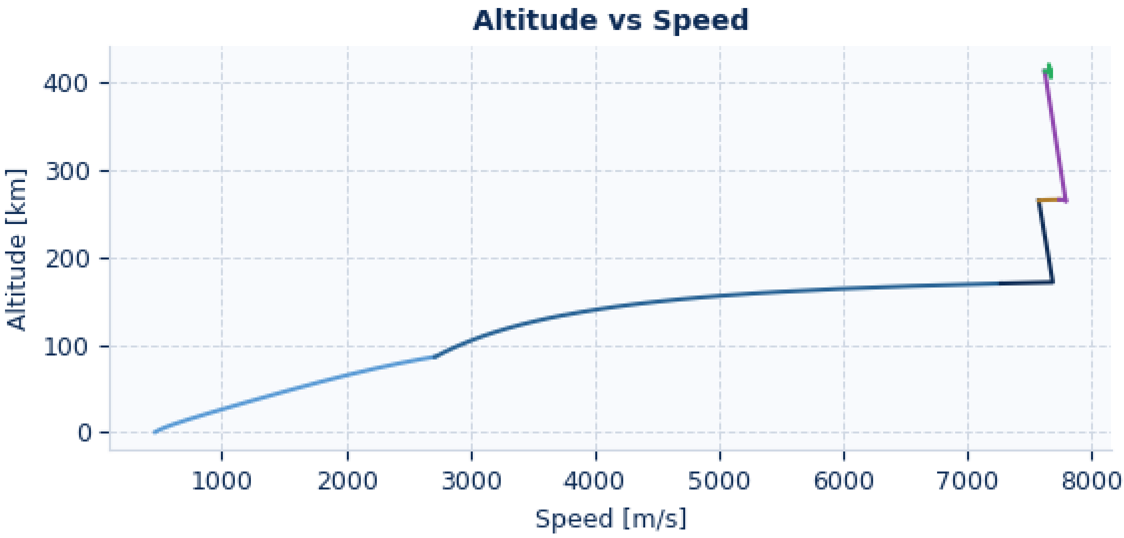


Figure 26: graph of altitude vs speed for our rocket

6. Appendix

Air density as a function of height:

```python
import numpy as np

# Atmospheric constants (International Standard Atmosphere)
rho0 = 1.225       # kg/m^3, sea level density
H = 8500.0         # m, scale height for exponential atmosphere
T0 = 288.15        # K, sea level temperature
L = 0.0065         # K/m, temperature lapse rate (up to 11 km)
R_air = 287.05     # J/(kg*K), specific gas constant for air
g0 = 9.80665       # m/s^2, standard gravity

def density_at_altitude(z):
    """
    Calculate air density using International Standard Atmosphere model.
    Returns density and atmospheric layer for altitude z in meters.
    """
    z_km = z / 1000.0

    if z_km < 11:
        layer = "Troposphere"
        T = T0 - L * z
        p = 101325 * (T / T0) ** (g0 / (L * R_air))
        rho = p / (R_air * T)

    elif z_km < 20:
        layer = "Lower Stratosphere"
        T = 216.65
        p_11 = 22632.1
        h = z - 11000
        p = p_11 * np.exp(-g0 * h / (R_air * T))
        rho = p / (R_air * T)

    elif z_km < 32:
        layer = "Upper Stratosphere"
        T = 216.65 + 0.001 * (z - 20000)
        p_20 = 5474.89
        h = z - 20000
        T_avg = (216.65 + T) / 2
        p = p_20 * np.exp(-g0 * h / (R_air * T_avg))
        rho = p / (R_air * T)

    elif z_km < 47:
        layer = "Mesosphere"
        T = 228.65 + 0.0028 * (z - 32000)
        p_32 = 868.02
        h = z - 32000
        T_avg = (228.65 + T) / 2
        p = p_32 * np.exp(-g0 * h / (R_air * T_avg))
```

```
        rho = p / (R_air * T)

    elif z_km < 51:
        layer = "Upper Mesosphere"
        T = 270.65
        p_47 = 110.91
        h = z - 47000
        p = p_47 * np.exp(-g0 * h / (R_air * T))
        rho = p / (R_air * T)

    elif z_km < 71:
        layer = "Lower Thermosphere"
        T = 270.65 + 0.0028 * (z - 51000)
        p_51 = 66.939
        h = z - 51000
        T_avg = (270.65 + T) / 2
        p = p_51 * np.exp(-g0 * h / (R_air * T_avg))
        rho = p / (R_air * T)

    elif z_km < 85:
        layer = "Upper Thermosphere"
        T = 270.65 + 0.002 * (z - 71000)
        p_71 = 3.9564
        h = z - 71000
        T_avg = (270.65 + T) / 2
        p = p_71 * np.exp(-g0 * h / (R_air * T_avg))
        rho = p / (R_air * T)

    else:
        layer = "Exosphere / Very Low Density Region"
        rho = 1.5e-5 * np.exp(-(z - 85000) / 6000)

    rho = max(rho, 1e-20)
    return {
        "altitude_m": z,
        "altitude_km": z_km,
        "layer": layer,
        "density_kg_m3": rho
    }


# Rangos definidos en el código, con inicio, fin y altitud de muestra (punto medio)
layers_info = [
    ("Troposphere",                      0,     11000,  5500),
    ("Lower Stratosphere",           11000,     20000, 15500),
    ("Upper Stratosphere",           20000,     32000, 26000),
    ("Mesosphere",                   32000,     47000, 39500),
    ("Upper Mesosphere",             47000,     51000, 49000),
    ("Lower Thermosphere",           51000,     71000, 61000),
    ("Upper Thermosphere",           71000,     85000, 78000),
    ("Exosphere / Very Low Density Region", 85000, None, 90000),
]
```

```
print(f"{'Rango (km)':>14} | {'Capa':<35} | {'Muestra (km)':>12} | {'Densidad
(kg/m³)':>18}")
print("-" * 90)
for layer_name, z_start, z_end, z_sample in layers_info:
    r = density_at_altitude(z_sample)
    rango = f"{z_start/1000:.0f} - {z_end/1000:.0f}" if z_end else
f"{z_start/1000:.0f} - inf"
    print(f"{rango:>14} | {layer_name:<35} | {z_sample/1000:>12.1f} |
{r['density_kg_m3']:>18.6e}")
```